Explaining Narrative Events with Generative AI and Structural Causal Models

Abstract. Interconnected narratives distribute explanations across actions, beliefs, resources and earlier decisions. A chronological summary can recover what happened while leaving unclear how these elements worked together. We present a method that combines generative AI interpretation with structural causal models to construct readable, inspectable causal maps. The generative component proposes events, contextual conditions, intermediate mechanisms and source-linked explanations. Restricted structural equations specify their joint implications, while partial functions preserve behavior that the narrative does not determine. Interventions then change selected events and propagate their consequences through fixed mechanisms. A worked Crow and the Pitcher example introduces the representation through contrasting contexts in which stone addition is essential or unnecessary. The principal application comprises six bounded Game of Thrones television cases. Detailed maps distinguish fuel from its deployment at Blackwater, preparation from attendance at the Red Wedding, encirclement from relief at Winterfell, and command from voluntary service at Astapor. Across 18 selected application questions, six distinguish the two retained scenario models, making consequential assumptions visible. The contribution is an integrated representation and construction workflow: readers can trace an interpretation to passages, inspect converging conditions, and compare hypothetical outcomes under explicit assumptions. The resulting maps complement narrative summaries with executable explanations whose evidential and functional commitments remain open to examination.

Keywords: narrative explanation, generative AI, structural causal models, causal maps, counterfactual reasoning

1 Introduction

An explanation of a narrative event often depends on more than its immediate predecessor. A battle can turn on an available resource, the decision to deploy it, a signal, an effective action and the opponent's exposure. A betrayal can depend on grievance, protection from retaliation and the victims' presence. These elements may appear in different scenes and involve different characters. Bringing them together requires interpretation; specifying how they work together requires a representation of their joint mechanism.

Generative AI offers a way to organize this material. It can identify event meanings, connect references across passages, propose intermediate steps and describe how one event creates an opportunity for another. A readable causal map preserves these proposals as inspectable relationships. Structural causal models (SCMs) add a complementary capability: they state how an event depends on its parents and on contextual conditions,

so that a hypothetical change has a defined consequence [8]. An arrow then carries both a narrative explanation and a role in a local mechanism.

This paper develops a workflow linking source passages, event definitions, causal descriptions, structural equations and intervention results. It has three complementary purposes: to connect passages to meaningful explanations; to combine informative graph annotations with explicit joint conditions; and to compare how intermediate events and alternative mechanisms contribute to a bounded outcome. The representation keeps observed events, proposed directional influences and assumptions about exact functions separate. This separation is especially valuable when a narrative depicts one course of events but leaves other possibilities open.

The principal application is the television narrative of *Game of Thrones*. Its interwoven character and faction stories connect earlier decisions to later opportunities and vulnerabilities. Alliances, grievances, beliefs, resources and timing interact, and local military events contribute to broader political consequences. Six bounded case studies organize these connections without attempting a single explanation of the entire series. Four detailed views show what the representation adds: Blackwater separates possession, deployment and ignition; the Red Wedding connects preparation with attendance; Winterfell exposes a consequential assumption about relief; and Astapor distinguishes authority, armed control and voluntary service.

We demonstrate source-linked maps and the execution of their declared mechanisms, retaining model provenance and unresolved alternatives. A worked example introduces the equations before the methodology and application. Supporting controlled results assess execution separately from the interpretation of the television narrative.

2 Motivating Example: The Crow and the Pitcher

A thirsty crow finds water below its reach, adds stones, raises the water and drinks. For a controlled reference example, an original retelling adds explicit rules about stone availability, an intact pitcher and initial reachability. These rules specify a small mechanism rather than relying on every reader to supply the same unstated physical assumptions.

Let S mean that enough stones are available, I that the pitcher is intact, and B that the water is initially reachable. Let A denote adding enough stones, R raising the water, T water being reachable after the action, and D drinking. Each variable is Boolean: one means that the scoped event occurs or condition holds. Sufficient water and the crow's readiness to act and drink are held fixed. The specified rules are

$$A = S, \qquad R = I \wedge A, \qquad T = I \wedge (B \vee R), \qquad D = T. \tag{1}$$

Here $\wedge$, $\vee$ and $\neg$ mean conjunction, disjunction and negation. Thus adding stones raises the water in the intact pitcher. Water is reachable either because it was reachable initially or because it has been raised, provided the pitcher retains access. Drinking follows reachability within the stated context. Figure 1 gives these rules readable event and relationship labels.

The factual inputs are $S = I = 1$ and $B = 0$. Executing the equations gives $A = R = T = D = 1$. Now prevent stone addition, keeping the water initially low. Replacing the first equation by $A = 0$ gives $R = T = D = 0$: the crow cannot drink through the

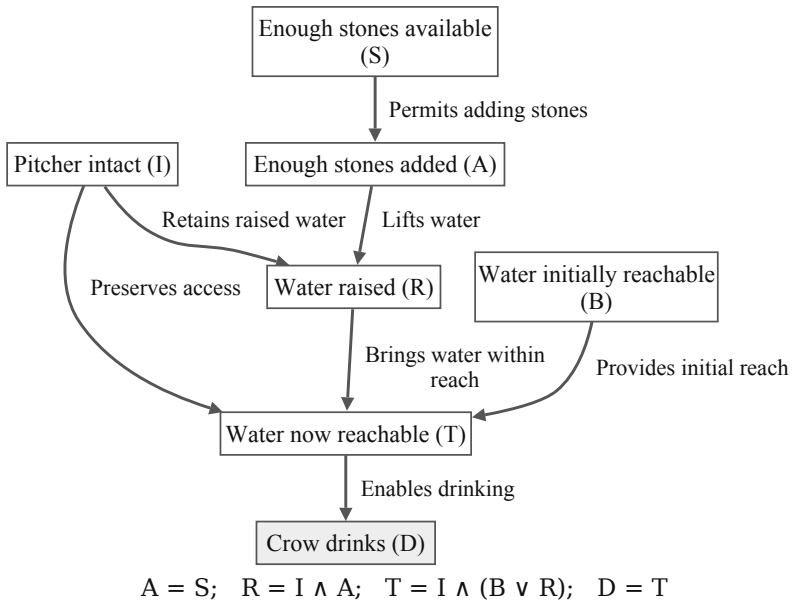


Fig. 1. Causal map of the Crow and the Pitcher, showing alternative routes to water reachability in the controlled story.

represented route. In a contrasting intervention, prevent addition and make the water initially reachable, setting $A = 0$ and $B = 1$. Then $R = 0$ but $T = D = 1$. The same prevented action has different consequences because the second route to reachability remains available.

The map therefore explains more than the sequence "stones, rising water, drinking." It distinguishes an action from its intermediate physical effect and separates that effect from an already-satisfied condition. Stones are required for drinking in the first context, but not in every context admitted by the model. The joint reachability equation makes this distinction precise. Additional controlled reference examples in Appendices A and B illustrate repair and credibility mechanisms.

3 Related Work

Narrative representations have long organized events around participants and temporal structure. Chambers and Jurafsky's narrative event chains connect events sharing a protagonist and learn partial temporal order [3]. Elson et al. extract literary social networks from dialogue interactions [4]. Such representations reveal event regularities and character relationships. The present maps make events, decisions, beliefs and resources explicit, and attach a joint mechanism to each modeled consequence. Knowing who interacts or which event tends to follow another is a useful starting point; an intervention additionally requires specifying what changes when an input is altered.

Recent work directly constructs causal graphs from narratives. Li et al. combine language-model summaries, linguistic features and iterative refinement to connect narrative events [7]. Koupaee et al. use semantic relation experts to construct causal event graphs for reasoning tasks [6]. These studies establish causal graph generation as an existing research direction. Our emphasis is the connection between explanatory relationships and executable local functions, including joint conditions and unresolved input configurations. A graph can state that two factors contribute to an event while leaving open whether they are substitutes or are both required; the structural layer exposes that choice.

Generative knowledge construction also motivates the division of labor. COMET uses pretrained representations to generate commonsense descriptions [1]. In narrative analysis, such knowledge can suggest a question or a missing intermediate step, but the resulting proposal must be checked against the selected source. Program-aided language models delegate calculation to an interpreter after generating an intermediate program [5]. We similarly separate interpretation from execution, while retaining the program as a source-linked explanatory object that a reader can inspect.

SCMs provide the semantics for interventions and for conclusions conditional on structural assumptions [8]. Bynum and Cho combine user-defined causal structure with language-model-defined mechanisms to generate observational and counterfactual data [2]. Our application instead compiles proposed narrative mechanisms into finite functions before querying them. The contribution lies in integrating semantic construction, readable mechanism records, partial specifications and fixed-function execution, rather than in introducing a new causal calculus.

4 Methodology

4.1 From passages to mechanism records

Construction begins by specifying an outcome, time interval and explanatory boundary. “The assault is repelled by the end of the battle” differs from “the enemy fleet suffers severe losses.” A useful variable states which event occurs, for whom, and by when. Its record includes an operational definition, domain, actor, event time, factual state and passage references. Reports and beliefs are distinguished from the external events they concern; the episode containing a retrospective report need not be the time of the reported event.

The source collection is broader than the initial model context. Episodes are organized into locatable passage containers. Retrieval first covers relevant actors, events and decisions; generation then proposes questions about missing connections. These questions search the broader case corpus, and surrounding passages are read before accepting a relationship. A proposed mechanism can therefore trigger further retrieval, splitting of a conflated variable, insertion of a supported mediator, correction of temporal scope or retraction of an unsupported connection. Chronological succession alone does not establish a causal edge.

Each proposed edge retains a short graphical phrase and a fuller process description. The record identifies conditions, qualifications, supporting passages, what each passage

supports, contradictory material and unresolved gaps. A relationship may be explicitly stated, contextually inferred, hypothetical, contradicted or unresolved. This description governs construction: a distinction between permission and action, for example, can require two variables before an equation is chosen. Children with several parents also receive a joint description of how those influences combine.

Three judgments remain separate. A passage can establish that an event occurred; it can support a directional influence; and it can constrain a particular functional form. Evidence for the first two does not automatically establish the third. "Helps" and "enables" need contextual interpretation, whereas "is necessary" or "exactly when" makes a stronger, testable commitment. Short labels communicate the process; the accompanying record carries its qualifications.

4.2 Explicit mechanisms and interventions

For each endogenous event or state X_i, the structural specification is

$$X_i = f_i(\mathrm{Pa}_i, U_i). \tag{2}$$

Here Pa_i is the collection of parent events represented in the graph, U_i contains declared contextual inputs, and f_i is the local mechanism. The present models use finite, principally Boolean domains and an acyclic dependency graph. The equations express scoped event occurrence or condition satisfaction. They are constructed rules, not statistical regression equations fitted to repeated observations.

An intervention $\mathrm{do}(X_k = x)$ replaces the equation or input assignment for X_k by the constant x. Other functions and the factual background remain fixed. For example, $\mathrm{do}(A = 0)$ in the Crow model replaces $A = S$ while retaining the rise, reachability and drinking rules. With $I = 1$ and $B = 0$, the change propagates through R and T to $D = 0$. A joint intervention replaces more than one assignment at once, as in $\mathrm{do}(A = 0, B = 1)$.

Several incoming arrows require a joint rule. In Eq. 1, the intact pitcher and stone addition are jointly required for the represented rise, whereas initial reachability and raised water provide alternative routes to reachability. Inhibition is represented by a negated condition, as when timely relief prevents an encirclement from persisting. An event can therefore contribute by supplying an input, completing a conjunction, providing a substitute route or removing an obstruction. Execution follows these mechanisms through intermediate events and convergence points. It evaluates a shared event once, even when several explanatory paths pass through it; the present application attaches neither probabilities nor additive responsibility weights to those paths.

4.3 Preserving incomplete mechanisms

A narrative commonly depicts only one input configuration. Unknown configurations remain unspecified until evidence or an explicit scenario assumption constrains them. For a child event, the implementation retains the family of functions compatible with its evidence constraints and declared assumptions. It can execute a complete scenario, enumerate a small compatible family, or report bounds obtained by a limited search. Appendix D gives the formal definition and the coverage distinction.

An important consequence is that factual agreement can coexist with counterfactual disagreement. Two functions can reproduce the same observed battle while assigning different roles to reinforcements. They are retained as alternative interpretations when the evidence does not settle the distinction. Frequencies of generated alternatives are not converted into probabilities. If only part of a family has been explored, agreement among those candidates does not by itself identify the answer.

4.4 Consistency, refinement and presentation

The workflow checks three links: source to description, description to function, and function to reported result. Source checking combines passage-location verification with contextual semantic review. For small finite mechanisms, formal claims can be checked over all input configurations. If a parent is asserted to be necessary, every configuration in which that parent is absent must make the child absent. Examining only the all-zero row is insufficient. A violated proposition yields a concrete counterexample. An unspecified row compatible with both outcomes instead identifies an encoding gap, unless the retained constraints exclude the prose claim altogether.

Discrepancies lead to bounded revision, rejection or an unresolved status, with original drafts retained. Established factual states cannot be relaxed simply to obtain an executable model. Results are recomputed from accepted functions, and explanatory prose receives the verified states and outcomes. Mechanical checks verify those values; semantic review considers interpretation and source support.

The retained application used Qwen3-14B-AWQ for source-grounded proposals and retrieval questions. A Codex assistant subsequently read explicit source packets and adjacent context, constructed the final mechanisms and scenario equations, and recorded revisions. These final application models are therefore assistant-mediated syntheses, rather than untouched outputs of the initial generator. At Blackwater, refinement separated Tyrion's signal from Bronn's ignition and inserted deployment between wildfire supply and damage. The stages used general modeling instructions, without a researcher-supplied Game of Thrones graph or fixed story equations.

Figures render accepted mechanism records. Solid edges denote explicitly supported directional relationships; dashed edges denote contextual interpretations. Neither style certifies an exact Boolean function. Compact equations state the scenario being examined, and fuller conditions and references remain in the mechanism catalogue. Detail views show a named local mechanism; conclusions about a later outcome use the complete retained model, whose additional mechanisms are described where needed. Mathematical symbols are local to each case.

5 Study Design and Narrative Material

5.1 Television case studies

The application corpus comprises 41 episode pages and 924 acquired scene containers from a single accessible transcription collection [9]. Containers can cover several locations and are treated as retrieval units, not inherently coherent causal episodes. Records

preserve episode identifiers, passage boundaries, source URLs and content hashes. Dialogue and stage descriptions are interpreted in their surrounding context. This study uses television continuity throughout; a later report can support an earlier event without becoming its cause.

Table 1 defines the six endpoints. They span mobilization, betrayal, battle outcomes and a bounded military departure. Their 69 variable records and 71 relationships include related events across cases; these are six interpretive case studies, not independent statistical samples. Each case retains a partial mechanism graph, two functionally distinct complete scenarios and three questions selected before computing their answers. The four main application figures were chosen for complementary explanatory content, while the retained portfolio covers every attempted case.

Table 1. Six bounded television outcomes and their principal explanatory mechanisms.

Case / endpoint	Scoped outcome	Principal mechanism
Stark–Lannister S01E07	Tywin orders Jaime's field deployment against the Riverlands	Allegation and detention connect to a dynastic response; a royal challenge is an alternative influence.
Northern coalition S01E09	Robb's host crosses the Twins with Frey support	Mobilization supplies a host; negotiated commitments secure permission and assistance.
Blackwater S02E09	Stannis's assault on King's Landing is repelled	A fire trap, a rear sortie and arriving relief provide interacting military routes.
Red Wedding S03E09	Robb's leadership and accompanying host are destroyed at the Twins	Concealed preparation and guest presence converge in attacks on the hall and camp.
Bolton campaign S06E09	The Stark coalition takes Winterfell from Ramsay	Encirclement, timely relief, pursuit and a gate breach connect the field battle to capture.
Astapor S03E04	Daenerys departs with freed Unsullied after breaking the masters' immediate armed control	Command transfer enables military action; freedom and the decision to remain define the departing force.

5.2 Supporting controlled material and resources

The three short-story checks use frozen natural retellings and explicitly marked controlled adaptations. The reference equations and four intervention answers per controlled story were specified before generation and withheld from the generator. Their reference figures illustrate the representation. The archived generations provide a separate assessment of semantic content and executable completeness, summarized in Appendix C.

An earlier controlled-text evaluation tests the execution component on 120 independently generated SCMs, split into 24 development, 24 calibration and 72 held-out

models. Complete rule texts and omissions of intermediate, outcome or multiple mechanisms provide 1,080 held-out questions. Model-level mean identified-intervention accuracy is 77.8% for synthesis followed by execution, versus 56.7% for direct answers; the paired difference is 21.0 percentage points (95% model-level bootstrap interval 9.6–32.8, 5,000 resamples). Sixteen failed syntheses remain in the primary denominator. On outcome-changing questions the corresponding accuracies are 80.0% and 54.6%. Factual carry-forward scores 86.2% overall because most identified questions preserve the outcome, but zero on changing questions. Single-model execution, alternative-model execution and the full explanation pipeline have identical identified-answer accuracy. This result supports executable reasoning over explicit rules, without attributing the gain to source weighting or richer explanations.

Qwen3-14B-AWQ ran on an NVIDIA RTX 4090 with 24 GB VRAM. The application source passes used 17 requests and 36,659 generated tokens: summed request time was 7.9 minutes, within 31.5 minutes of inference-service residency. The simple-story run used 18,814 generated tokens and 6.26 minutes of service residency. These timings exclude the subsequent interpretive and editorial work. The application assistant's exact served build and token usage were unavailable; the archived records identify that generation route separately.

6 Game of Thrones: Results and Interpretation

6.1 Blackwater: a resource becomes an implemented trap

The first question is how possession of wildfire contributes to damage inflicted on Stannis's fleet. The relevant sequence spans Tyrion's acquisition of the resource in S02E05 and its use in S02E09. Figure 2 distinguishes control of the supply from deploying fuel in the bay. It also separates Tyrion's signal from Bronn's physical action. The phrases "Supplies fire trap," "Signals the shot" and "Ignites the trap" identify different processes rather than assigning the explosion to one undifferentiated action.

Let W denote wildfire supply under Tyrion's control, T deployed fuel, S his signal, I Bronn's ignition, B fleet exposure and D severe fire losses. The displayed scenario uses $T = W$, $I = S$ and $D = T \wedge I \wedge B$. Its background holds the deployment decision, prepared archer and absence of substitute ignition fixed. The conjunction describes a particular coordinated trap: fuel supplies the combustible material, ignition activates it, and exposed ships are the objects damaged. Resource possession alone does not bypass the implementation steps.

This local outcome is distinct from repelling the assault. The adjacent battle material shows that Stannis survives and continues the attack. The full map therefore includes Tyrion's rear sortie and the later relief force. In both retained scenarios, preventing Bronn's ignition removes the depicted fire losses but leaves the final assault repelled when the other represented actions occur. Removing relief separates the scenarios: one requires relief together with a holding action, whereas the other permits naval attrition and the sortie jointly to substitute. These are alternative completions of the battle mechanism, not conclusions established by the spectacle of the explosion. The map makes it possible to trace a local tactical contribution without equating that contribution with the whole victory.

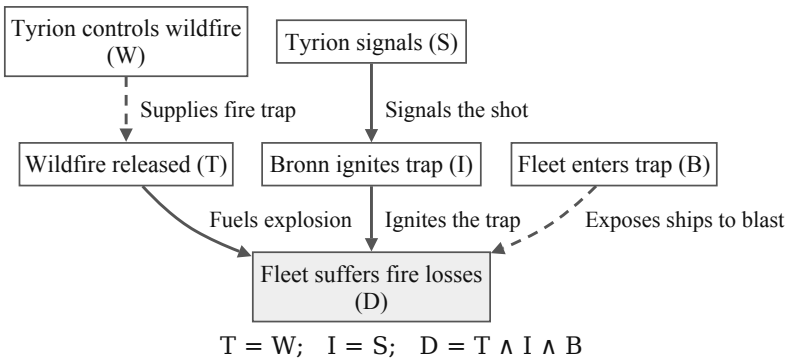


Fig. 2. Fuel deployment, ignition and fleet exposure combine in the Blackwater wildfire mechanism, separating preparation from implementation and immediate damage.

6.2 The Red Wedding: preparation meets attendance

The Red Wedding asks how a grievance and an opportunity for attack become a coordinated ambush. The selected material connects the replacement-marriage arrangement in S03E06, the wedding in S03E09, and the retrospective discussion of protection in S03E10. In Fig. 3, the latter supports an interpretation of earlier assurances; it is not placed later in the causal sequence merely because it is reported later.

Let G be Walder Frey's grievance, A Tywin's assurances, T prepared attackers, N acceptance of the replacement marriage, H the promise of guest safety, P the coalition's presence at the wedding, and X the hall ambush. One scenario specifies $T = G \wedge A$, $P = N \wedge H$ and $X = T \wedge P$. A second replaces the preparation rule by $T = A$. The first interpretation makes grievance and protection complementary; the second allows strategic protection to sustain preparation without the represented grievance.

The edge about risk concerns *Frey's perceived risk when betraying Robb*. Assurances can reduce anticipated retaliation and thereby support preparation. They are distinct from the guest-safety promise that encourages the intended victims to remain. Likewise, a grievance can motivate retaliation without itself placing the victims within reach. "Prepares concealed attack" and "Puts leaders within reach" converge on the ambush, exposing the interaction between capability and opportunity.

Removing grievance prevents the final modeled destruction in the first scenario but not the second. Removing assurances prevents it in both. These answers depend on the declared preparation rules and on fixed downstream assumptions about attack success. The complete case connects the hall ambush and attack on the camp to the destruction of Robb's leadership and accompanying host; it does not equate that bounded outcome with the disappearance of all northern resistance. The explanatory gain is the separation of motivation, perceived protection, attendance and implementation, each of which can be examined against different passages.

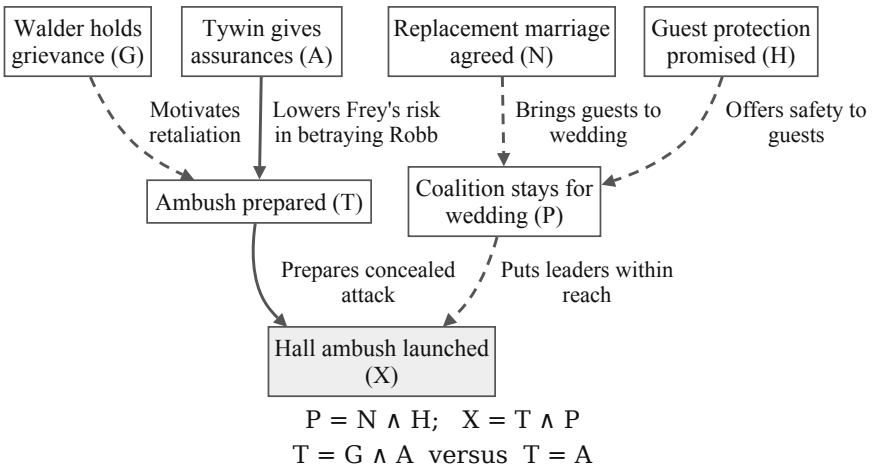


Fig. 3. Prepared attackers and guest attendance converge on the Red Wedding ambush; grievance and perceived protection have distinct roles.

6.3 Winterfell: relief and an obstruction to advance

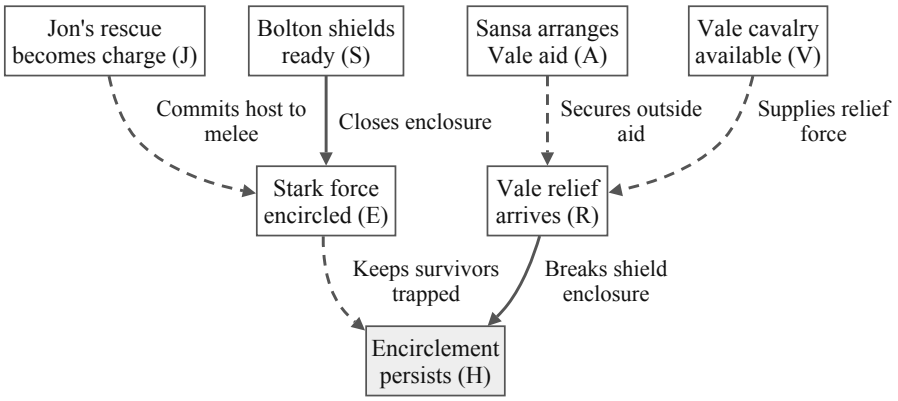
At the Battle of the Bastards, the outcome is the capture of Winterfell by the end of S06E09. Figure 4 links the field mechanism to three intervention answers. Its local variables distinguish Jon's exposed advance J, a ready Bolton shield force S, encirclement E, Sansa's arrangement of assistance A, available Vale cavalry V, timely relief R, and encirclement persisting into the advance stage H. The rules are $E = J \wedge S$, $R = A \wedge V$ and $H = E \wedge \neg R$.

The last equation expresses inhibition: relief removes an obstruction. It does not mean that the earlier encirclement never occurred. That temporal distinction preserves both the formation of the trap and its subsequent disruption. Evidence for the battle comes from S06E09; the interpretation of Sansa's earlier arrangement also uses surrounding material and the retrospective discussion in S06E10.

The scenario comparison concerns what happens next. Let G denote an available Stark field force and P its effective pursuit and advance. The *relief-required* scenario uses $P = G \wedge R \wedge \neg H$; the *untrapped-advance* scenario uses $P = G \wedge \neg H$. In the full model, a gate breach K requires both P and an available Wun Wun W, so $K = P \wedge W$; capture Y follows $Y = K \wedge P$. All these later mechanisms remain fixed between interventions within each scenario.

Removing timely relief alone leaves the encirclement persistent and prevents capture in both scenarios. Removing Rickon's use as bait alone prevents the modeled exposed advance and still permits capture. The joint intervention removes both bait and timely relief. Under the retained bait-to-advance rule, no trap forms, so $H = 0$. The relief-required model nevertheless gives $P = 0$ and no capture; the untrapped-advance model gives $P = 1$ and capture, with the other inputs unchanged. The different answers pinpoint the unresolved question: is relief required for effective advance in its own right, or only

(a) Encirclement and relief



(b) Starks take Winterfell?

	Relief required	Untrapped advance
No Vale relief	No	No
No Rickon bait	Yes	Yes
Neither bait nor relief	No	Yes

Fig. 4. Encirclement and relief at Winterfell: (a) the local mechanism; (b) intervention outcomes under two assumptions about the subsequent advance.

to remove this obstruction? A chronological account depicts relief arriving; the map states exactly what extra assumption makes it necessary in an unobserved situation.

6.4 Astapor: authority, force and voluntary service

The Astapor case follows the bargain in S03E03 through the departure in S03E04. Its outcome combines the breaking of the masters' immediate armed control with departure of a freed force. Figure 5 separates three relationships often compressed into a single claim that Daenerys obtains an army: negotiated authority, use of armed force and a subsequent decision to remain.

Let O be the dragon offer, B the accepted bargain and C command transfer. In panel (a), $B = O$ and $C = B$ retain the counterparties' observed acceptance and implementation as scenario assumptions. In panel (b), Q is the order against the masters and U defeat of their guards, with $U = C \wedge Q$. Authority provides access to an obedient force, while the order directs that force toward the particular action. Keeping these nodes separate distinguishes having command from exercising it.

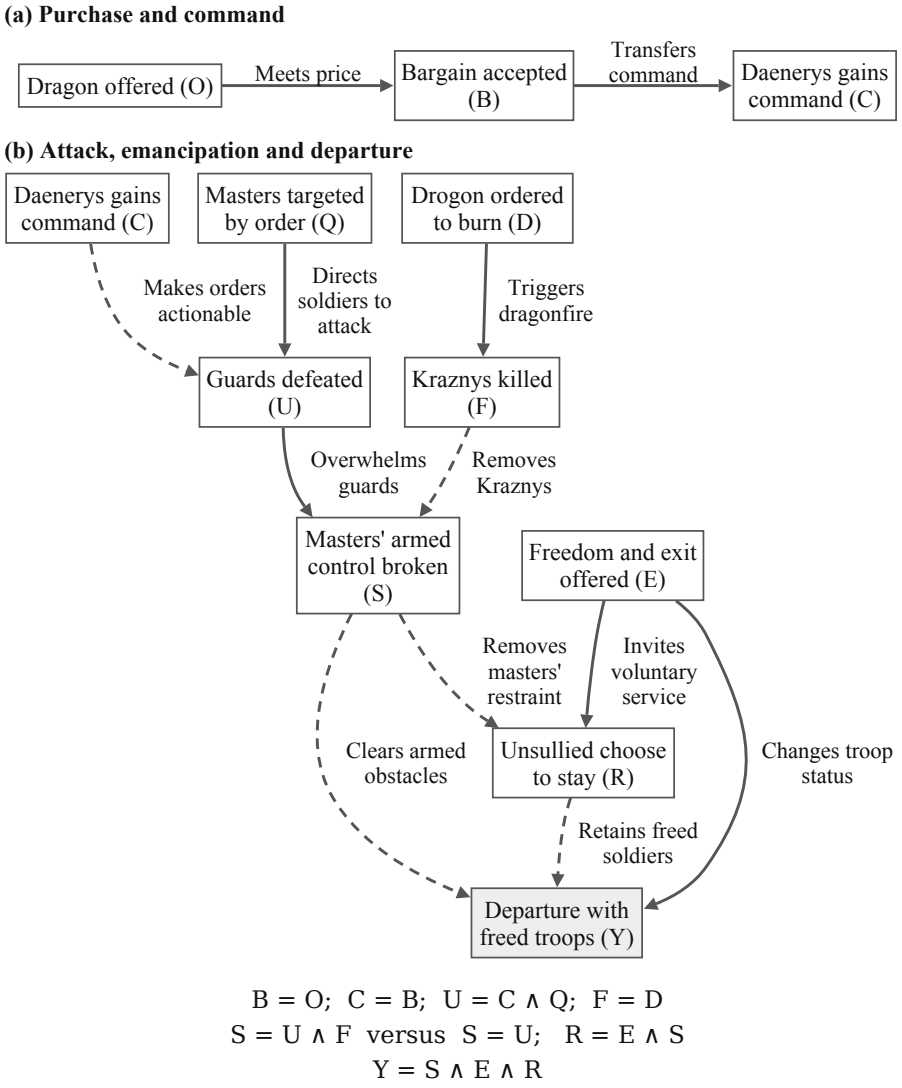


Fig. 5. The Astapor sequence connects (a) purchase and command transfer with (b) armed action, emancipation and departure with freed troops.

The parallel dragon route distinguishes the Dracarys order D and the killing of Kraznys by fire F, using $F = D$. Let S denote the masters' immediate armed control being broken. The combined-force scenario specifies $S = U \wedge F$; the alternative specifies $S = U$. Both reproduce the represented events. Preventing the fire strike therefore separates them, whereas withholding the order against the masters prevents the departure outcome in both. The causal question is whether the successful ground action would suffice within this boundary, not whether the depicted dragon attack occurred.

Finally, let E be emancipation with permission to leave, R the troops' decision to remain, and Y departure with a freed army. The retained completion uses $R = E \wedge S$ and $Y = S \wedge E \wedge R$. The first rule assumes the observed willingness to stay persists when freedom and immediate security are present; it is not a general law that emancipation produces loyalty. The distinct decision node preserves the meaning of voluntary service even in this simplified deterministic scenario. The outcome excludes subsequent governance of Astapor. The map thus traces how institutional authority enables military action while making freedom and continued service separate commitments.

6.5 What the portfolio establishes

Across the six cases, the maps distinguish immediate actions from resources, permissions, beliefs and longer-lived conditions. The northern coalition case, for example, keeps assembling a force separate from securing its crossing and Frey support. The Stark–Lannister case records an allegation about the dagger as information acting on decisions, rather than treating the allegation's content as established fact. These distinctions make different points in an explanation available for inspection.

All 18 selected application questions were executed in both explicit scenarios; six, one per case, have different outcomes across those scenarios. The remaining twelve agree within the selected pair. The broader partial families admit both outcome values for every question. This combination is informative: the complete scenarios show the consequences of particular assumptions, while the partial representation records that the observed narrative does not select those assumptions uniquely. Agreement between two scenarios is consequently a limited sensitivity result. The figures are readable views of these mechanisms, not a claim that the equations recover a uniquely true account of the stories.

7 Discussion and Practical Applications

The principal practical value is an inspectable chain from a passage to a relationship, from that relationship to a joint rule, and from the rule to a hypothetical consequence. At Blackwater, a reader can ask whether a proposed explanation concerns fuel, deployment, ignition or victory. At the Red Wedding, the reader can distinguish protection offered to the betrayer from safety promised to the guests. At Winterfell, the comparison identifies the advance rule responsible for two different answers. At Astapor, the map keeps command and continuing allegiance conceptually separate.

This organization suggests uses in teaching, literary analysis and the examination of narrative accounts. A reader could compare competing interpretations at a particular

mechanism instead of debating an entire plot summary. An editor could inspect whether a claimed consequence has an intervening action or merely follows it in time. A classroom exercise could change one contextual condition and explain why the same action matters differently. These are illustrated capabilities and proposed applications; reader performance has not been measured.

The short-story reassessment clarifies the roles of interpretation and formalization. The initial controlled Humpty and Wolf outputs correctly expressed all their reference joint rules in prose, and Crow expressed three of four. They also left substantial executable behavior unspecified, and the Wolf output misstated a factual response. Thus useful semantic construction can coexist with incomplete mathematical encoding and a genuine factual error. The direct baseline answered all twelve controlled questions correctly, while no first-attempt generated model met the full executable acceptance criterion. The reference maps and the final refined application maps serve different evidential roles, which the study design keeps explicit.

The study uses one dependent transcription family and assistant-refined application models; its mechanism judgments await independent researcher adjudication. Boolean contrasts and fixed backgrounds simplify continuous, strategic and institutional processes. The results support conditional scenario comparison, not estimated causal probabilities or general narrative understanding. A next evaluation should adjudicate event, directional and functional support separately, then test whether readers can locate the assumption responsible for a changed answer. This would assess explanatory capability beyond visual preference or prose length.

8 Conclusion

Generative interpretation and structural modeling provide complementary parts of a narrative explanation. The first supplies meaningful events, contextual distinctions and candidate mechanisms; the second makes their joint implications executable and inspectable. The Crow example shows how an action's importance depends on an alternative route. Six Game of Thrones cases extend this representation to interacting decisions, resources, opportunities and political consequences. Their maps distinguish mechanisms that a chronological summary can compress, and their scenario comparisons expose assumptions that matter to hypothetical outcomes. The resulting contribution is a practical method for turning interconnected narrative material into causal maps that readers can trace, question and revise while retaining a clear relationship between evidence, interpretation and computation.

Declaration on Generative AI

Generative AI assisted drafting, editing, coding, figure preparation and consistency checking. Its scientific use is described in Section 4. This draft awaits the authors' substantive review and revision. Before submission, the authors must verify and take responsibility for the arguments, interpretations, results and final text.

References

1. Antoine Bosselut, Hannah Rashkin, Maarten Sap, Chaitanya Malaviya, Asli Celikyilmaz, and Yejin Choi. COMET: Commonsense transformers for automatic knowledge graph construction. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4762–4779. Association for Computational Linguistics, 2019.
2. Lucius E. J. Bynum and Kyunghyun Cho. Language models as causal effect generators. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 2096–2115. Association for Computational Linguistics, 2025.
3. Nathanael Chambers and Dan Jurafsky. Unsupervised learning of narrative event chains. In *Proceedings of ACL-08: HLT*, pages 789–797. Association for Computational Linguistics, 2008.
4. David K. Elson, Nicholas Dames, and Kathleen R. McKeown. Extracting social networks from literary fiction. In *Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics*, pages 138–147. Association for Computational Linguistics, 2010.
5. Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 10764–10799. PMLR, 2023.
6. Mahnaz Koupaee, Xueying Bai, Mudan Chen, Greg Durrett, Nathanael Chambers, and Niranjan Balasubramanian. Causal graph based event reasoning using semantic relation experts. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 26169–26199. Association for Computational Linguistics, 2025.
7. Zehan Li, Ruhua Pan, and Xinyu Pi. Beyond LLMs: A linguistic approach to causal graph generation from narrative texts. In *Proceedings of the 7th Workshop on Narrative Understanding*, pages 36–51. Association for Computational Linguistics, 2025.
8. Judea Pearl. Causal inference in statistics: An overview. *Statistics Surveys*, 3:96–146, 2009.
9. ScreenScripts. Game of Thrones: Episode transcripts. `https://screenscripts.com/game-of-thrones/`. Television continuity; accessed 13 October 2026.

A Humpty Dumpty: Damage and Repair

The controlled reference retelling specifies a fall and damage, followed by an unsuccessful attempt by the king's helpers. It does not identify Humpty as an egg. Let F denote falling, D initial damage, A a repair attempt, E availability of an effective method, R successful repair and Y remaining damage. The added rules are

$$D = F, \qquad R = D \wedge A \wedge E, \qquad Y = D \wedge \neg R. \tag{3}$$

The factual inputs $F = A = 1$ and $E = 0$ give $D = Y = 1$ and $R = 0$. Figure 6 separates trying to repair from having an effective method and actually succeeding. With the attempt retained, $\mathrm{do}(E = 1)$ gives $R = 1$ and $Y = 0$. Jointly making a method available and preventing the attempt gives $R = 0$ and $Y = 1$. Availability and implementation are complementary in this adaptation.

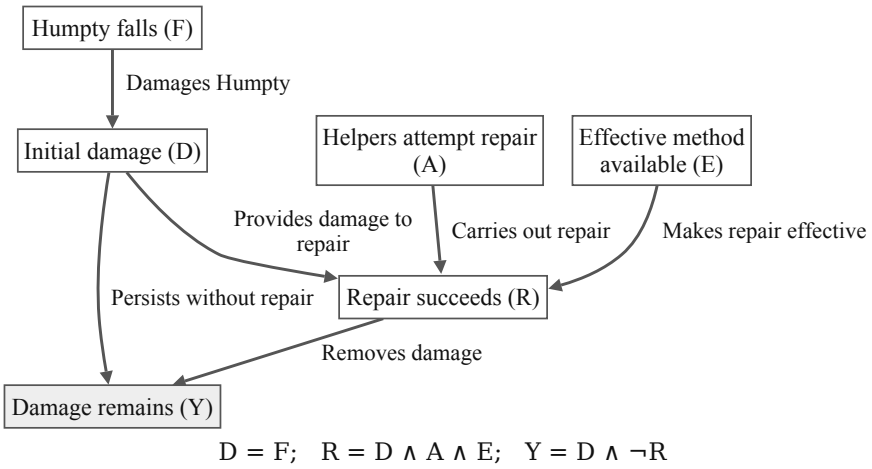


Fig. 6. Damage, attempted repair and effective repair remain distinct in the controlled Humpty Dumpty story; successful repair removes remaining damage.

Failure by the depicted helpers establishes an unsuccessful repair at the endpoint, not the impossibility of every possible repair. The controlled sentences introduce the explicit success rule; the natural retelling leaves that counterfactual open. Keeping attempted repair, effective repair and remaining damage distinct prevents the observed failure from silently becoming permanent irreparability.

B The Boy Who Cried Wolf: Warning and Credibility

This controlled reference retelling distinguishes earlier false alarms from a new warning about a real wolf. Let F denote the earlier false alarms, L loss of credibility, W a real wolf appearing, N the new warning, V timely village response and H harm to the flock. The adaptation explicitly stipulates

$$L = F, \qquad N = W, \qquad V = N \wedge \neg L, \qquad H = W \wedge \neg V. \tag{4}$$

The background fixes the opportunity to warn and the effectiveness of timely help. With $F = W = 1$, the boy warns, credibility is absent, villagers do not respond in time, and the flock is harmed. The boy's death is not an event in the specified story.

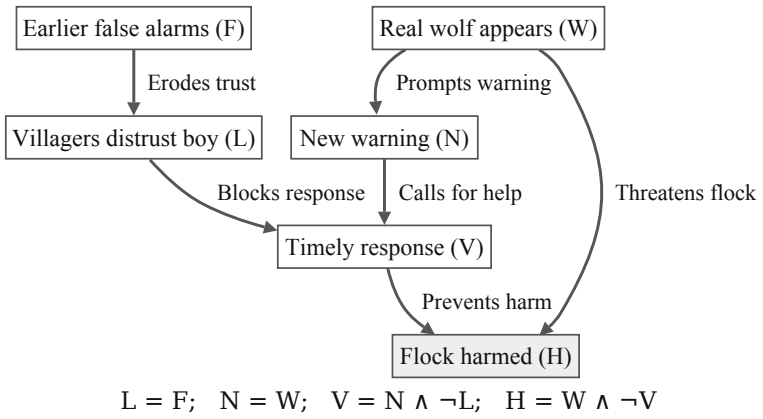


Fig. 7. Credibility and a new warning jointly determine timely response in the controlled Wolf story; timely response prevents flock harm.

Preventing the earlier false alarms, do($F = 0$), restores credibility and yields timely help, so $H = 0$. But jointly preventing false alarms and the new warning, do($F = 0, N = 0$), gives $V = 0$ and $H = 1$. Figure 7 therefore keeps disbelief distinct from silence: a credible boy who does not warn provides no trigger for the stipulated response. Guaranteed prevention of harm belongs to the adaptation's added rule, rather than being inferred from the natural tale's unhappy outcome.

C Assessment of the Short-Story Generations

None of the six first-attempt models met strict executable acceptance, including after structural normalization. All contained relevant events and relationships. Table 2 separates supported factual assertions and correctly expressed rules from accepted execution. These assistant reassessment judgments concern semantic correspondence, not literal wording. The three stories, rather than their individual questions or variables, are the descriptive units.

Table 2. Semantic content and executable completeness of the unmodified controlled-story generations.

Story	Supported facts	Correct prose rules	Accepted model	Direct answers
Humpty	6/6	3/3	No	4/4
Wolf	5/6	4/4	No	4/4
Crow	8/8	3/4	No	4/4

The controlled Humpty generation correctly says that repair succeeds exactly when damage, an attempt and an effective method coincide. Its alternative variable names are harmless; an edge making damage a prerequisite for attempting repair needs substantive clarification because the stated prerequisite concerns success. The Wolf generation distinguishes distrust, a new warning, response and harm and correctly describes their joint rules, but records timely response as occurring when the controlled story says it does not. The Crow generation correctly distinguishes initially reachable water from water raised by stones, but adds an intact-pitcher prerequisite to stone addition itself. That changes intermediate behavior when stones are available and the pitcher is not intact, even though the drinking outcome remains zero in that configuration.

Across eleven controlled joint descriptions, nine are compatible with but not entailed by their partial encodings. Two, both in Wolf, conflict with the wrongly recorded response state. Missing encoding in an explicitly specified story is therefore different from uncertainty in its source. A post hoc diagnostic projection retaining supplied rows finds two of Crow's four answers determined and correct, with the other two unresolved. It leaves all four Humpty questions unresolved; Wolf includes one definitely wrong answer. These projections are separate partial-model analyses, not replacements for the original unavailable executed answers. All twelve direct answers and all twelve author-reference executions match the frozen reference answers.

The natural outputs capture useful qualitative content: Humpty's fall and attempted repair, Wolf's loss of credibility and ignored warning, and Crow's rise-to-reach-to-drink sequence. Their supported factual counts are 4/6, 8/8 and 8/8 respectively. Natural Humpty introduces unsupported irreversible damage, whereas the other two preserve gaps about protective help and a leaking pitcher. Equivalent terminology and short causal phrases are assessed in context. Explicit necessity or sufficiency claims still require the quantified behavior they assert. No substantive correction to an archived generation is counted as an initial generation success.

D Partial Functions and Coverage

For a fixed graph and finite domains, let E be the retained evidence and A the declared assumptions. If $C_i(E)$ and $C_i(A)$ are the corresponding local constraints, the admissible functions for event i are

$$\mathcal{F}_i(E, A) = \{f_i : f_i \text{ satisfies } C_i(E) \text{ and } C_i(A)\}. \tag{5}$$

Let $\mathcal{M}(E, A)$ contain the combinations consistent with all established factual states. For a query q and fixed factual background u^*, the outcome set for target Y is

$$\mathcal{Y}(q) = \{Y_{M,q}(u^*) : M \in \mathcal{M}(E, A)\}, \tag{6}$$

where $Y_{M,q}(u^*)$ is obtained by replacing only the equations targeted by q in model M. A singleton identifies the answer within this declared family only when exhaustive evaluation or a valid bound establishes that no other outcome is possible. It says nothing about omitted variables or alternative boundaries.

The broad application families fix observed input–output configurations and leave other rows unknown. They may include locally inactive parent links despite qualitative directional support; they are conservative computational envelopes, not complete translations of everything the passages imply. Each retained application mechanism has at most three parents, allowing all local Boolean tables to be considered. Global products range from 65,536 to 8,388,608 functions, so joint search is bounded at 512 candidates. Every reported broad outcome set is nevertheless certified as $\{0, 1\}$: compatible witnesses attain both values of the declared binary target domain. A complete enumeration is unnecessary to establish that particular set. The two named complete scenarios per case add stronger assumptions and are not claimed to exhaust this family.

For formal prose checking, necessity of a positive input requires output zero in every row where that input is zero; sufficiency requires output one in every row where it is one. Exact conjunction and inhibition assertions require their complete stated truth conditions. The retained targeted mutations change a conjunction to disjunction, remove a necessary input or reverse inhibition, and the corresponding quantified checks detect the contradictions. An informal statement that a factor helps does not receive one of these stronger translations without an explicit interpretive justification.